Event-Based Visual-Inertial State Estimation for High-Speed Maneuvers

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***Abstract*—Neuromorphic event-based cameras are bioinspired visual sensors with asynchronous pixels and extremely high temporal resolution. Such favorable properties make them an excellent choice for solving state estimation tasks under high-speed maneuvers. However, failures of camera pose tracking are frequently witnessed in state-of-the-art event-based visual odometry systems when the local map cannot be updated timely or feature matching is unreliable. One of the biggest roadblocks in this field is the absence of efficient and robust methods for data association without imposing any assumptions on the environment. This problem seems, however, unlikely to be addressed as in standard vision because of the motion-dependent nature of event data. To address this, we propose a map-free design for event-based visual-inertial state estimation in this article. Instead of estimating camera position, we find that recovering the instantaneous linear velocity aligns better with event cameras' differential working principle. The proposed system uses raw data from a stereo event camera and an inertial measurement unit (IMU) as input, and adopts a dual-end architecture. The front-end preprocesses raw events and executes the computation of normal flow and depth information. To handle the temporally nonequispaced event data and establish association with temporally nonaligned IMU's measurements, the back-end employs a continuous-time formulation and a sliding-window scheme that can progressively estimate the linear velocity and IMU's bias. Experiments on synthetic and real data show our method achieves low-latency, metric-scale velocity estimation. To the best of the authors' knowledge, this is the first real-time, purely event-based visual-inertial state estimator for high-speed maneuvers, requiring only sufficient textures and imposing no additional constraints on either the environment or motion pattern.**

***Index Terms*—Event-based cameras, simultaneous localization and mapping, visual-inertial fusion, visual odometry.**

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I. Introduction

As opposed to standard cameras that capture synchronously the absolute brightness at a fixed frame rate, event cameras are bioinspired sensors that acquire visual information in the form of a stream of asynchronous per-pixel intensity changes (called "events"). Endowed with micro-second temporal resolution, event cameras can sense at exactly the same rate as the scene dynamics (induced by either ego motion or independent moving objects). Consequently, they do not suffer from motion blur and are qualified for state estimation tasks involving aggressive ego motion. The main challenge of building such an event-based visual state estimator is to solve in real time the tracking and mapping subproblems by exploiting photometric and geometric constraints encoded in the generative model of events.

Although event-based methods for camera pose tracking [1], [2] have been proved effective in tackling aggressive ego-motion estimation, most event-based visual odometry (VO) systems are not always qualified. Early works using a monocular event-based camera (e.g., [3], [4]) require a very gentle motion (typically a local-loopy behavior) for the initialization of a local 3-D map, based on which the camera pose can be tracked using a 3D–2D registration pipeline. To remove such a limitation on the initialization, the authors in [5] and [6] further used a stereo event-based camera to improve the efficiency and accuracy of mapping. However, tracking failure is still witnessed when the ego motion of the camera suddenly becomes violent (mainly in terms of the angular velocity). This is because the mapping subproblem solver cannot update the local map in time.

To circumvent this issue, researchers resort to visual-inertial fusion and feature-based methods. Although the event-inertial combination seems not as complimentary as the standard-visual-inertial counterpart, it is still beneficial to fuse event data with measurements of an IMU. First, an IMU's measurements can be used as a motion prior to propagate the estimated state. Moreover, such motion prior can be leveraged to obtain a motion-compensated image of warped events (IWE), on top of which a feature-based pipeline can be simply built [7]. Obviously, the success of [7] largely relies on a sharp IWE, which is obtained via a geometric 3D–2D warping operation that requires the median depth of the local map. This warping operation, however, can easily fail if there are relatively large depth variations. To alleviate the front-end's dependence on motion prior, some works [8], [9] straightforwardly establish feature association on simple

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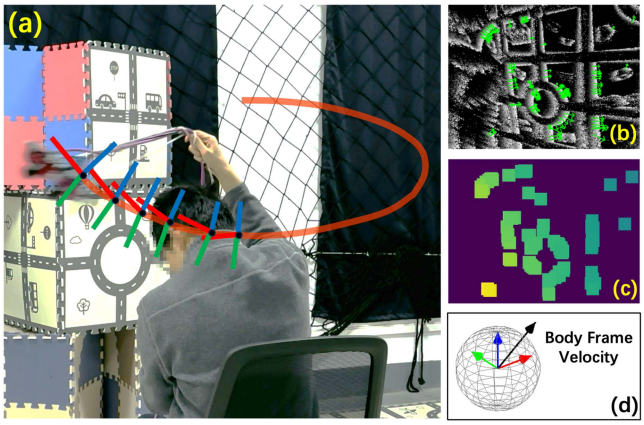


Fig. 1. Proposed system takes as input the event data from a stereo event-based camera and inertial measurements from an IMU, and recovers the instantaneous linear velocity of the sensor suite. (a) High-speed motion induced by horizontally spinning the tethered sensor suite. (b) Sparse and asynchronous normal-flow estimates from event data. (c) Sparse depth estimates at the corresponding origins of the normal-flow vectors. (d) Illustration of the resulting instantaneous linear velocity in the sensor's body frame.

image-like representations of events (e.g., time surfaces [10]). These ad hoc solutions may work to some extent, but the success of large-baseline feature matching is hardly guaranteed. This is due to the fact that edges parallel to the epipolar line hardly trigger any events, and thus, junction-shape patterns may not be observed completely. Consequently, the appearance similarity evaluated on such naive image-like representations could be violated in a sudden variation of linear velocity.

Noticing the dependence on the first-order kinematics, the authors in [11] and [12] proposed a set of speed-invariant representations, which lead to constant thickness of edge patterns irrelevant to camera speed. Although we have witnessed a lot trials on event-based feature detection and tracking [12], [13], [14], [15], most of them either cannot give satisfied feature tracking results as their standard-vision counterparts [16], [17] do, or are computationally inefficient for real-time applications. A few works [18], [19], [20] manage to establish efficiently event-to-edge association, based on which parallel tracking and mapping is achieved under aggressive motion at an ultra-frame rate. However, these solutions are only applicable in the presence of man-made geometric patterns. All these issues drive us to think about a question: Is there a more rational design of state estimation for an event-based visual-inertial system that is more consistent with the differential working principle of event cameras? To answer this, we propose a first-order-kinematic state estimator, namely a velometer, using a stereo event-based camera and an IMU (see Fig. 1). Considering the special working principle of event cameras, we propose a continuous-time pipeline for the linear velocity estimation problem. Specifically, our approach exploits the generative model of event-based normal flow induced by the first-order kinematics of the event camera. Besides, the dynamic constraint from the IMU is also utilized. The contribution of this article is summarized as follows.

1) A novel design of state estimator for an event-based visual-inertial system, which aligns well with the differential working principle of event cameras and recovers the first-order kinematic state (i.e., linear velocity) by fusing event data and inertial measurements. Combined with gyroscope measurements, full 6-degree of freedom (DoF) pose estimation can be achieved through dead reckoning with only a single integration step, providing a novel approach for state estimation tasks during high-speed maneuvers.
2) A rigorous derivation for computing normal flow from the spatial–temporal profile of event data. In addition, we propose a method to evaluate normal-flow uncertainty by deriving the variance of the normal-flow magnitude from event data statistics, which reflects the flatness of the locally fitted spatio-temporal plane.
3) A continuous-time framework (i.e., B-spline regression) for event-based visual-inertial fusion, which can handle asynchronous event measurements and establish data association with temporally nonaligned measurements from the accelerometer. Prior information from historical estimates is constantly incorporated into the current B-spline regression through a marginalization operation within a sliding-window framework.
4) A comprehensive evaluation using qualified data, in which we compare our method against several visual-event-inertial-based counterparts. To do this, we also provide a synthetic dataset and a real one collected using various configurations and mobile platforms. All the data feature high-speed maneuvers commonly seen in practical applications of robotics and computer vision.

This work has evolved from our paper [21], and the technical improvements are clarified at the end of Section II.

Outline: The rest of this article is organized as follows. First of all, a literature review focusing on the topic of event-inertial state estimation is given in Section II. Then we provide a clear problem statement and an overview of the proposed system in Section III. To better explain our method, we revisit several key concepts as preliminaries in Section IV. Our method is discussed in Section V, followed by the experimental evaluation in Section VI. Finally, Section VII concludes this article.

II. Related Work

Event-based state estimation intends to recover the ego motion and a 3-D map (optional) by exploiting specific constraints on motion and scene geometry encoded in the generative model of event data. Due to the special working principle and sensor characteristics, event cameras appear as a complementary sensor in some visual-inertial systems aiming to address challenging scenarios that involve high-speed maneuvers and/or high-dynamic-range (HDR) illumination. We also realize the existence of a large body of literature on the topic of event-based motion estimation, such as the optimization-based methods for geometric model fitting on event data [22], [23], [24], [25], [26], [27], and learning methods that recover motion and structure [28], [29]. To be compact, our literature review focuses on three aspects: 1) A brief history of event-based VO ranging from primitive pipelines to modern advancements; 2) Fusion of event

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TABLE I
LITERATURE REVIEW ON EVENT-BASED VISUAL (-INERTIAL) ODOMETRY

Reference	Sensors	Motion	Limitations	Real-Time Performance
Weikersdorfer et al. [30]	Monocular	3-DoF (X-Y-Yaw)	Requires straight-line landmarks on a plane	✓
Kim et al. [31]	Monocular	3-DoF (Orientation)	–	✗
Gallego et al. [22]	Monocular	3-DoF (Orientation)	–	✗
Kim et al. [3]	Monocular	Local-Loopy 6-DoF	Requires GPU resources	✓
Rebecq et al. [4]	Monocular	Local-Loopy 6-DoF	–	✓
Zhu et al. [32]	Monocular + IMU	Local-Loopy 6-DoF	Assumes landmarks are planar in 3-D	✗
Zhou et al. [5]	Stereo	Normal 6-DoF	–	✓
Niu et al. [6], [33]	Stereo + IMU	Normal 6-DoF	–	✓
Chamorro et al. [19]	Monocular	Normal 6-DoF	Requires straight-line landmarks	✓
Wang et al. [34]	Stereo	Normal 6-DoF	–	✗
Liu et al. [35]	Stereo + IMU	Normal 6-DoF	–	✓
Ghosh et al. [36]	Stereo	Normal 6-DoF	–	✓
Wang et al. [37]	Stereo + IMU	Normal 6-DoF	–	✓
Mueggler et al. [38]	Monocular	Aggressive 6-DoF	Requires a known planar pattern	✗
Rebecq et al. [7]	Monocular + IMU	Aggressive 6-DoF	Assumes a small variation in local depth	✓
Vidal et al. [39]	Monocular (E+F) + IMU	Aggressive 6-DoF	Assumes a small variation in local depth	✓
Chen et al. [9]	Stereo (E+F) + IMU	Aggressive 6-DoF	Assumes a small variation in local depth	✓
Xu et al. [40]	Monocular + IMU	Aggressive 6-DoF	Requires straight-line landmarks	✗
Lu et al. [21]	Stereo + IMU	Aggressive 6-DoF	–	✓
This work	Stereo + IMU	Aggressive 6-DoF	–	✓

data and IMU measurements; and 3) Different choices of state variables in the problem of event-based state estimation.

A. *History of Event-Based Visual (-Inertial) Odometry*

The development of event-based VO can be seen as a process of gradually overcoming the challenges in handling full 6-DoF motion and even aggressive maneuvers, while reducing assumptions and constraints on the scene. To be focused, our discussion is restricted to existing event-based full-VO (VIO) systems that do not rely on a prior 3-D map, as listed in Table I.

To the best of the authors' knowledge, the earliest work of event-based VO can date back to 2013 and it is validated on a vacuum cleaner-like ground robot equipped with a monocular event camera looking upward [30]. The proposed algorithm can generate high-quality 2-D map of the ceiling while precisely tracking the robot's 3-D motion, including the 2-D translation and the yaw component. Afterward, the general 3D-rotation estimation problem is studied and conquered in [22], [31]. Kim et al. [31] proposed a probabilistic filtering-based method that can jointly estimate the camera's 3-D rotation while reconstructing an intensity gradient map. Differently, Gallego and Scaramuzza [22] leveraged the scheme of contrast maximization to regress the instantaneous 3-D angular velocity of an event camera.

The first system that can handle 6-DoF motion is seen in [3]. Based on the constant-brightness assumption in the log intensity domain, Kim et al. [3] proposed a direct method that implements three interleaved probabilistic filters to solve the subproblems of mapping, camera pose tracking, and additionally, recovering the log intensity information. Subsequently, Rebecq et al. [4] proposed a pure geometric method to justify that recovering the intensity information is not mandatory. The proposed mapping approach determines the 3-D location of structures by searching the maximum ray intersection in the disparity space image (DSI), and the camera motion is estimated through a 3D–2D registration process, in which the resulting 3-D map is aligned to the synthesized 2-D event map. Contemporaneously, Zhu et al. [32] presented the first event-based VIO system based on an event data association scheme for robust feature tracks. These three pioneering pipelines are, however, limited by the requirements of gentle loopy behaviors in the initialization and slow expansion of the local map. Hence, none of their success has been justified on data collected using a mobile platform.

To handle normal 6-DoF motion, Zhou et al. [5] presented the first event-based VO pipeline (ESVO) using a stereo event camera. The method exploits spatio-temporal consistency of the events across the image planes of the cameras to solve both localization and mapping subproblems of VO. To enhance ESVO's mapping efficiency and tracking accuracy, Niu et al. [6], [33] proposed an IMU-aided version of ESVO, which scales better with modern event cameras of VGA resolution and overcomes ESVO's degeneracy issue of camera pose tracking. Similarly, building on top of ESVO and incorporating an IMU is also witnessed in [35] and [37], while other follow-ups like [36] further enhance the mapping accuracy via a correspondence-free mapping module that estimates depth by maximizing ray density. In addition, several indirect methods are also proposed, such as [19] and [34], which use line and point features respectively to achieve parallel mapping and positioning under normal 6-DoF motion. Last but not the least, the capability of handling aggressive maneuvers is only seen in pipelines that fuse event data with an IMU's measurements [7], [9], [21], [40], which are discussed in detail in Section II-B.

It is worth mentioning that some works highlight the versatility and adaptability of event cameras when combined with complementary sensing modalities. Weikersdorfer et al. [41] pioneered the integration of depth-augmented dynamic

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vision sensors for 3-D simultaneous localization and mapping (SLAM). Following their work, Zuo et al. [42] further demonstrated enhanced robustness in challenging conditions, such as low-light and high-speed scenarios. Integrating event cameras with frame-based cameras, Kueng et al. [43] introduced a low-latency indirect method by extracting features from frames and tracking them in the event domain. Applying a similar strategy, Mahlknecht et al. [44] further explored event camera-based odometry for planetary robots, highlighting its potential in extreme environments. To overcome the challenge of appearance changes in indirect methods, Hidalgo-Carrio et al. [45] proposed a direct approach. Their method leverages the generative model of events in the front end to fuse frames and event data, while the back end employs a photometric bundle adjustment to refine poses and depth by minimizing normalized brightness errors.

We also note that there is limited research exploring deep learning-based approaches for event-based VO, particularly those aimed at achieving enhanced performance on recent benchmarks. By tailoring a novel deep patch selection mechanism [46] to event data, Klenk et al. [47] presented deep event VO, the first deep monocular event-only system. To further integrating frames, Pellerito et al. [48] proposed the first end-to-end learned image- and event-based VO system, which leverages novel recurrent, asynchronous, and massively parallel encoders specifically designed to fuse asynchronous event streams with traditional image data. These methods pioneer the use of end-to-end learning-based architectures but are limited to offline systems, necessitating additional engineering efforts for practical applications. Therefore, we exclude them from our evaluation benchmark.

B. Event-Based Visual-Inertial Odometry

There is a large body of literature on this topic, most of which are built on top of a discrete-time back-end that fuses events and IMU measurements in a manner of either probabilistic filtering or nonlinear optimization. These methods differ mainly in the front tend, where event-based data association is established. Early works [7], [32] detect corners on an edge map generated by naively accumulating a set of events onto the image plane. To keep tracking those corners, a two-step expectation-maximization method [32] is proposed to estimate the event-based optical flow and update the feature template. However, the computational complexity is so high that few successive reports apply this strategy in the front-end. With prior knowledge of short-term relative motion and depth information of the local map, Rebecq et al. [7] leveraged the idea of motion compensation [23] to restore sharp IWEs, on which corner features are constantly tracked. More recently, an increasingly popular trend (e.g., [9]) is to apply traditional image-based feature detection and tracking methods on a certain image-like representation (e.g., [10], [11]) of event data. Despite their relative successes, the front-end of these ad hoc methods are not as theoretically sound as their standard-vision counterparts due to the motion-dependent nature of event data. Moreover, the publicly available datasets [49], [50], [51], [52], while sometimes captured using an agent moving at high linear velocities, generally do not include instances of significant large-displacement optical flow, making them less qualified for assessing state-estimation performance under high-speed maneuvers.[1] The superior performance of these methods lies in the improvement of robustness under low-texture or HDR conditions. Few of them (except for [7]), however, have ever demonstrated the ability to solve state estimation problems under aggressive maneuvers.

A relatively smaller body of literature looks into the design of the back-end that can deal with asynchronous and temporally nonaligned data. The employment of a continuous-time framework to the event-inertial fusion is first reported in [53], which parameterizes the camera trajectories in special euclidean group (SE)(3) with cubic B-splines. Based on the generative model of event and inertial observations, the proposed probabilistic approach seeks the optimal estimation of the posteriori of the camera pose over a time interval, given the local map, event data, and IMU measurements. The limitations of this method include, on one hand, the requisite of a known 3-D map. On the other hand, it does not manifest the ability to address aggressive ego motion estimation. An alternative way to implement a continuous-time framework is using Gaussian process. Wang and Gammell [34] represented a continuous-time trajectory with states defined at different timestamps of event feature tracklets. By using a white-noise-on-acceleration motion prior, the smoothness of camera trajectories is enforced considering an agent's physical plausibility. The overall trajectory is estimated using nonparametric Gaussian process regression. Still, it does not manifest effectiveness under aggressive motion.

C. First-Order Kinematics: A Better Choice?

Its' only recently that researchers began to consider the question: What is the optimal choice for state variables in event-based state estimation? Most existing event-inertial fusion pipelines target absolute camera poses. However, this choice is inconsistent with the differential nature of event data. To this end, Xin et al. [54] proposed the first event-based state estimation method on the level of first-order kinematics. The method relies on trifocal tensor geometry, which exploits the relationship between line features and first-order kinematics of the event camera. With known angular velocity, up-to-scale linear velocity can be estimated in a closed form. A successive work [40] is presented recently. It extends [54] by a direct solution of the linear velocity, and furthermore, delivers a tight event-inertial fusion back-end achieving more reliable estimation of linear velocity. These methods demonstrate their feasibility in tackling the problem of aggressive ego-motion estimation. However, a limited scope of usage is clearly seen because of the dependence on straight lines, which are only available in man-made structural environments. Besides, real-time performance is not demonstrated in [40]. We also note that a very recent breakthrough [55] demonstrates the feasibility of estimating full-DoF egomotion (both angular and linear velocities) without relying on additional sensor measurements or motion priors. However, it remains constrained

[1] Refer to our definition of high-speed maneuvers in Section III.

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to scenes containing long straight lines and cannot operate in real-time.

Considering the limitation of existing methods, we propose a novel state estimator at the level of first-order kinematics, namely a velometer, using a stereo event-based camera and an IMU. Different from feature-based methods, our approach leverages a low-level observation, namely, event-based normal flow, which is directly induced by the first-order kinematics of the event camera. To be compatible with the asynchronous property of event data, we also employ a continuous-time framework using cubic B-splines to parameterize state variables. The resulting pipeline is a map-free velocity estimator that tightly fuses event-based normal flow and IMU measurements, which can estimate linear velocity under aggressive motion in real time. Furthermore, we demonstrate in Section VI-C that the proposed linear velocity estimator allows for recovering position information with only one integration operation, leading to more accurate dead-reckoning results. Consequently, our method outperforms other methods in the sense that it can handle state estimation tasks under aggressive 6-DoF ego motion without imposing specific restrictions on the environment (except for the presence of enough textures) and computational resources.

Remark: The system proposed in this article is an enhanced version of our previous work [21] with the following theoretical supplements and engineering efforts.

1) We use the theorem of uncertainty propagation to derive the covariance of normal flow (reflecting the flatness in the local spatio-temporal profile), which can be used to reweight the normal-flow residual term in the objective function [see (17)].
2) We introduce a sliding-window scheme with a marginalization operation, which applies the prior information onto adjacent B-spline control points and suppresses potential oscillations in the resulting linear velocity estimates due to noisy measurements.
3) We present a more extensive evaluation, using additional self-collected data, to demonstrate the superior performance of our system in solving state estimation tasks under high-speed maneuvers.

III. System Overview

Problem statement: Let $\mathcal{X}^{\mathtt{A}}$ denote a state $\mathcal{X}$ described in the coordinate system $\mathtt{A}$; the state $\mathcal{X}$ can be linear velocity $\mathbf{v}$ or angular velocity $\boldsymbol{\omega}$. The relative pose between two coordinate systems (e.g., the world frame $\mathtt{W}$ and the body frame $\mathtt{B}$) is denoted by a rigid transformation $\mathbf{T}_{\mathtt{B}}^{\mathtt{W}} \doteq \{\mathbf{R}_{\mathtt{B}}^{\mathtt{W}}, \mathbf{t}_{\mathtt{B}}^{\mathtt{W}}\}$, which consists of a rotation $\mathbf{R}_{\mathtt{B}}^{\mathtt{W}}$ and a translation $\mathbf{t}_{\mathtt{B}}^{\mathtt{W}}$. We assume that the stereo event-based camera has been precalibrated, including the spatial–temporal calibration between the left and right event cameras, and between the left camera and the IMU. For simplification, we further assume that the left camera's coordinate system coincides with that of the IMU, which is denoted as the body coordinate system ($\mathtt{B}$). Given as input the raw events from a stereo event camera and the measurements from an IMU, the goal is to estimate instantaneous linear velocity $\mathbf{v}^{\mathtt{B}}$ of the sensor suite in metric scale. The environment is assumed to be static, consistent with the assumptions commonly made in most SLAM systems.

Regarding high-speed maneuvers: The aim of this work is to address state estimation tasks in high-speed scenarios, where motion blur often degrades or entirely disrupts frame-based solutions. Therefore, the definition of *high-speed maneuvers* should address the primary causes of motion blur and focus on factors commonly encountered in practical applications of high-speed mobile robots. These factors include not only the ego-motion pattern of the observer but also the scene attributes. Specifically, our definition is made on the following two factors.

1) A general 6-DoF motion pattern, where both linear and angular velocities can be large enough to generate significant large-displacement optical flow.
2) A proper motion parallax (i.e., an appropriate distance between the scene and observer) that avoids two extreme cases: 1) gentle maneuvers near structures that may still cause motion blur; and 2) high linear velocities (e.g., vehicle motion) in large-scale environments that are insufficient to induce motion blur.

Factor 1) distinguishes our defined scenarios from existing works (e.g., [56]), which demonstrate successful tracking of aggressive local shaking movements of an event camera, provided the local depth remains within the field of view. Such a shaky motion pattern is, however, rarely to occur in practical mobile robot applications. Notably, factor 2) helps further refine the scope of the definition. If the scene depth is relatively large, motion blur is hardly to occur even if the observer undergoes translational motion at high linear velocity. On the other hand, motion blur can be induced when the observer is close to the scene, even if the observer's motion is not particularly fast. In addition, excessively small scene depth can lead to significant distortion in visual information, adversely affecting the quality of event data association.

System design: As seen in Fig. 2, the entire system adopts a dual-end architecture. The front-end computes the depth and normal flow using as input the events from a stereo event camera. Specifically, the depth is estimated by exploiting the stereo association established on a widely used image representation of events, i.e., time surfaces [10], as discussed in Section IV-C; the normal flow (including its uncertainty information) at selected event points is computed from the spatio-temporal profile of events using the method presented in Sections V-A and V-B. The resulting depth and normal flow, plus the IMU preintegration results, are passed to the back-end.

The back-end estimates the linear velocity by fusing the event data and the IMU's measurements. To deal with the temporally nonaligned data from the two heterogeneous sensors, we employ a continuous-time parameterization, jointly optimizing the linear-velocity parameters and IMU bias (see Section V-C). To initialize the continuous-time nonlinear optimization problem, a robust linear solver in provided and presented in Section V-D. Besides, the marginalization method is adopted to keep the size of the sliding window constant, while preserving the historical information as priors in the marginalized linear system (see Section V-E). The ultimate result consists of the estimated instantaneous linear velocity and the updated IMU bias, where the latter is back-propagated to the preintegration function in the front-end.

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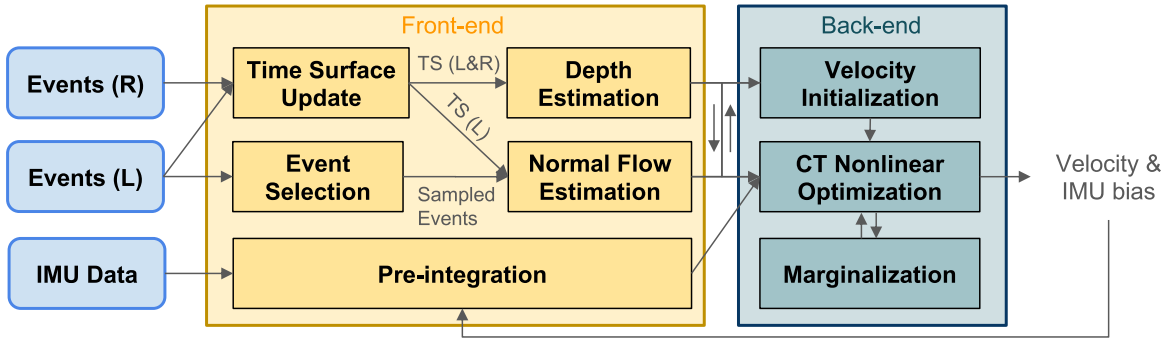


Fig. 2. Flowchart of the proposed event-based visual-inertial velometer system. The system takes as input the events from a stereo event camera and an IMU's inertial measurements, and reports the estimated linear velocity and updated IMU's bias.

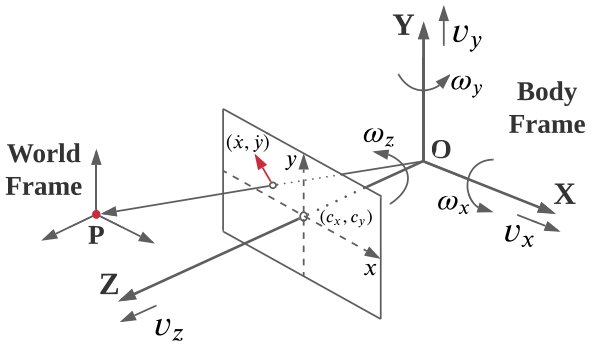


Fig. 3. Geometry and kinematics involved in the problem of motion flow. An ideal perspective camera model is used to illustrate that the 3-D point $\mathbf{P}$ is projected on the image plane. The components of linear velocity and angular velocity are marked along their corresponding axis. The resulting motion flow is denoted by the red vector $(\dot{x}, \dot{y})$.

IV. Preliminaries

Before explaining our method, we need to revisit several essential concepts as preliminaries, including *motion flow*, *normal flow*, and *event-based depth estimation*.

A. Motion Flow

Motion flow, also known as optical flow, refers to derivatives of a 2-D image location w.r.t time. It describes how a 2-D point would traverse on the image plane. As illustrated in Fig. 3, the motion flow of a textured 3-D point $\mathbf{P}^{\mathrm{B}} = [X, Y, Z]^{\mathrm{T}}$ can be determined according to the famous equation

$$\begin{aligned}\begin{bmatrix}\dot{x}(t)\\ \dot{y}(t)\end{bmatrix} &= \begin{bmatrix}\frac{v_z x' - f v_x}{Z} + \frac{\omega_x}{f}x'y' - \omega_y\left(f + \frac{x'^2}{f}\right) + \omega_z y'\\ \frac{v_z y' - f v_y}{Z} + \omega_x\left(f + \frac{y'^2}{f}\right) - \frac{\omega_y}{f}x'y' - \omega_z x'\end{bmatrix}\\ &= \frac{1}{Z(\mathbf{x})}\mathbf{A}(\mathbf{x})\mathbf{v}^{\mathrm{B}}(t) + \mathbf{B}(\mathbf{x})\boldsymbol{\omega}^{\mathrm{B}}(t)\\ \mathbf{A}(\mathbf{x}) &= \begin{bmatrix}-f & 0 & x'\\ 0 & -f & y'\end{bmatrix}\\ \mathbf{B}(\mathbf{x}) &= \begin{bmatrix}\frac{x'y'}{f} & -\left(f + \frac{x'^2}{f}\right) & y'\\ f + \frac{y'^2}{f} & -\frac{x'y'}{f} & -x'\end{bmatrix}\end{aligned} \tag{1}$$

where $(\dot{x}, \dot{y})$ denotes the motion flow vector at pixel $\mathbf{x} \doteq (x, y)$, $\boldsymbol{\omega}^{\mathrm{B}} = [\omega_x, \omega_y, \omega_z]$ the angular velocity, $\mathbf{v}^{\mathrm{B}} = [v_x, v_y, v_z]$ the linear velocity, and f the focal length. The relative image coordinates w.r.t the optical centre (c_x, c_y) are denoted by x' and y', namely, $x' \doteq x - c_x$ and $y' \doteq y - c_y$.

The motion flow (1) is first disclosed in [57]. It is clearly demonstrated that the motion flow is uniquely determined by the 3-D information (Z) and the first-order kinematics of the camera ($\mathbf{v}^{\mathrm{B}}$ and $\boldsymbol{\omega}^{\mathrm{B}}$). In other words, given the angular velocity $\boldsymbol{\omega}^{\mathrm{B}}$ (e.g., obtained from an IMU's measurements) at a time instant, the linear velocity $\mathbf{v}^{\mathrm{B}}$ can be estimated if the motion flow and depth information at no fewer than two pixels are known.

B. Normal Flow (NF)

For standard vision, the motion flow is typically determined according to the *Horn–Schunck* model [58], which assumes that the brightness of a point remains constant when observed from two spatial and temporal neighboring perspectives. By expanding the image brightness function with first-order Taylor expansion, the brightness-constancy constraint goes as

$$I(\mathbf{x} + d\mathbf{x}, t + dt) \approx I(\mathbf{x}, t) + \nabla_{\mathbf{x}} I(\mathbf{x}) \begin{bmatrix}dx\\ dy\end{bmatrix} + \nabla_t I(\mathbf{x}) dt. \tag{2}$$

This equation provides only one constraint such that only the partial component parallel to the image gradient direction can be determined as

$$\dot{\mathbf{x}}_n(t) = -\frac{\nabla_t I(\mathbf{x})}{\|\nabla_{\mathbf{x}} I(\mathbf{x})\|^2} \nabla_{\mathbf{x}} I(\mathbf{x}) \tag{3}$$

where $\dot{\mathbf{x}}_n$ denotes the projection of the flow $\dot{\mathbf{x}}$ on the image gradient direction $\nabla_{\mathbf{x}} I$. This is referred to as the well-known *aperture problem*, and the partial component $\dot{\mathbf{x}}_n$ is typically called *normal flow*.

Let $\mathbf{n}$ denote the image gradient direction at $\mathbf{x}$, namely $\mathbf{n} = \frac{\nabla_{\mathbf{x}} I(\mathbf{x})}{\|\nabla_{\mathbf{x}} I(\mathbf{x})\|}$. By multiplying both sides of (1) with $\mathbf{n}^{\mathrm{T}}$, we obtain

$$\|\dot{\mathbf{x}}_n(t)\| = \frac{1}{Z(\mathbf{x})} \mathbf{n}^{\mathrm{T}} \mathbf{A}(\mathbf{x})^{\mathrm{B}} \mathbf{v}(t) + \mathbf{n}^{\mathrm{T}} \mathbf{B}(\mathbf{x})^{\mathrm{B}} \boldsymbol{\omega}(t). \tag{4}$$

Since normal flows rather than full motion flows can be estimated straightforwardly from raw events, (4) is used as a constraint in the estimation of linear velocity.

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C. Event-Based Depth Estimation

Existing methods of recovering depth information from event data can be classified into two categories. The first category is called the "temporal stereo" method, in which a monocular event-based camera is often used. Temporal stereo methods assume the camera's motion to be known as a prior [59], [60]. They utilize events occurred over a temporal window to determine the 3-D location of structures by searching the maximum in the DSI through either voting in the discrete space [59], [60] or continuous optimization [23]. The second category is called the "instantaneous stereo" method, which typically applies a stereo event camera that have been calibrated extrinsically and synchronized temporally [61], [62], [63]. Typically, this kind of methods consists of two steps: 1) Searching corresponding events occurred in the left and right event cameras; 2) Triangulation: To enhance the ratio of true-positive matching, state-of-the-art methods [64], [65] utilize a hybrid metric to measure the similarity of two events, which jointly considers temporal coherence, epipolar constraint, and motion consistency in the spatio-temporal neighborhood. Since depth estimation is not the focus of this work, we simply apply the instantaneous stereo matching method (i.e., block matching on time surfaces [10]) implemented in [5].

V. Methodology

In this section, we first introduce our method of event-based normal flow estimation (Section V-A), followed by deriving the uncertainty of normal-flow magnitude from the local spatio-temporal profile (see Section V-B). Second, we present a continuous-time approach that fuses the two heterogeneous measurements by progressively fitting a cubic B-spline in the space of linear velocity (see Section V-C). To provide a reasonable initial value for the nonlinear optimization problem, moreover, we provide a robust linear solver for the initialization (see Section V-D). Finally, we employ a sliding-window scheme for keeping as constant the number of optimized B-spline control points, where we marginalize the oldest control point out and apply the prior information on the last few control points remained in the sliding window (see Section V-E).

A. Event-Based NF Estimation

To estimate visual flows from raw events, Benosman et al. [66] proposed a local differential approach that fits local spatio-temporal planes on the time surface composed of coactive events. The time surface is defined specifically by the function $\mathcal{T}(x, y)$, which returns the timestamp by which the most recent event occurred at pixel $[x, y]^{\mathrm{T}}$. The spatio-temporal plane's gradient is calculated as the temporal derivative w.r.t the pixel coordinate, namely, $\nabla_{\mathbf{x}}\mathcal{T} \doteq \left[\nabla_x\mathcal{T}, \nabla_y\mathcal{T}\right]^{\mathrm{T}}$, which reports the timestamp difference between spatially adjacent events in the direction of x and y, respectively.

Let us assume that the brightness gradient direction is orthogonal to edges. As shown in Fig. 4(a) and (b), the time surface $\mathcal{T}$ can be regarded as a temporal profile induced by the normal flow $\dot{\mathbf{x}}_n$, and we have

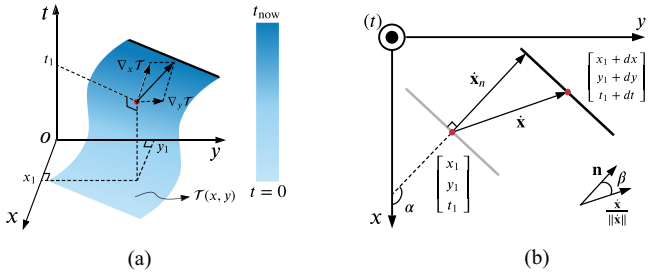


Fig. 4. General principles of normal flow estimation from raw event streams. (a) Time surface $\mathcal{T}(x, y)$ is spanned in the spatio-temporal domain as the edge (black) traverses-temporal flow estimation as spatio-temporal planar fitting. The temporal flow $\nabla_{\mathbf{x}}\mathcal{T}$ can be estimated by fitting a local plane. A color scale for the temporal values is provided aside. The darker the color, the closer the time is to the current moment. (b) Normal flow $\dot{\mathbf{x}}_n$, namely, the component of the motion flow $\dot{\mathbf{x}}$ along the direction of brightness gradient $\mathbf{n}$, is parallel to the temporal flow $\nabla_{\mathbf{x}}\mathcal{T}$—trigonometry in regard to motion flow $\dot{\mathbf{x}}$, normal flow $\dot{\mathbf{x}}_n$, and brightness gradient direction $\mathbf{n}$.

$$\begin{bmatrix}\nabla_x\mathcal{T}\\ \nabla_y\mathcal{T}\end{bmatrix} = \frac{1}{\|\dot{\mathbf{x}}_n\|}\begin{bmatrix}\cos(\alpha)\\ \sin(\alpha)\end{bmatrix}. \tag{5}$$

In fact, only normal flows can be recovered from temporal derivatives due to the aperture problem [β is unknown in Fig. 4(b)], and thus, the visual flow estimated by [66] does not refer to the full motion flow $\dot{\mathbf{x}}$. The method in [66] is partially correct in the sense that the temporal gradient reveals the normal flow's direction. The reciprocal of the temporal gradient's components would not straightforwardly give correct normal flows. The amplitude of the normal flow can be determined using (5), and the normal flow vector is calculated as

$$\dot{\mathbf{x}}_n = \|\dot{\mathbf{x}}_n\|\mathbf{n} = \frac{1}{\sqrt{\nabla_x\mathcal{T}^2 + \nabla_y\mathcal{T}^2}}\mathbf{n} \tag{6}$$

where $\mathbf{n} = \frac{\nabla_{\mathbf{x}}\mathcal{T}}{\|\nabla_{\mathbf{x}}\mathcal{T}\|}$ refers to the direction of the normal flow.[2] We will justify our derivation by comparing against the result of [66] in the ablation study (see Section VI-E). Note that Valeiras et al. [67] have also pointed out the incorrect method of normal flow computation in [66] and provided the correct formula.

B. Uncertainty of Event-Based NF

Accurate estimates of normal flow are key to the subsequent event-IMU fusion. However, the estimation accuracy depends on the quality of the temporal profile of events, which may be degraded by sensor noise. Therefore, a way is needed to assess the uncertainty of the normal-flow estimates. To this end, we derive the variance of normal-flow magnitude from the event data statistics that reflect the flatness of the local plane fitted in the spatio-temporal domain.

Given a set of coactive events in the neighborhood (denoted by $\mathcal{E} \doteq \{e_i\}_{i=1}^{N}$, where each event $e_k = (x_i, y_i, t_i, p_i)$ is a tuple consisting of the event location (x_i, y_i), the timestamp t_i, and the polarity p_i), the local plane fitting task in the spatio-temporal domain aims to determine the coefficients of the plane equation

[2] Note that we abuse the notation by duplicating $\mathbf{n}$, because the direction of the normal flow is parallel to that of the local image gradient.

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$(t = ax + by + c)$ by solving the following least-square problem:

$$a^\star, b^\star, c^\star = \arg\min_{a,b,c} \|\mathbf{X}\mathbf{p} - \mathbf{t}\|^2 \tag{7}$$

where $\mathbf{t} = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix}$, $\mathbf{X} = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots \\ x_N & y_N & 1 \end{bmatrix}$, and $\mathbf{p} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Using the known theory of linear regression [68], the covariance matrix of the coefficients can be derived analytically

$$\mathrm{Cov}(\mathbf{p}) = \frac{\sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}}{N - 3} \tag{8}$$

where $\sigma^2 = \sum_{i=1}^{N} (t_i - c - ax_i - by_i)^2$. Substituting the resulting coefficients into (5), we have

$$\|\dot{\mathbf{x}}_n\| = \frac{1}{\sqrt{a^2 + b^2}}. \tag{9}$$

The detailed derivation of (9) can be found in Appendix A. Furthermore, applying the uncertainty propagation method in [69], the first-order approximation of the normal-flow amplitude's variance is

$$\mathrm{Var}(\|\dot{\mathbf{x}}_n\|) \approx \frac{\partial \|\dot{\mathbf{x}}_n\|}{\partial \mathbf{p}} \mathrm{Cov}(\mathbf{p}) \frac{\partial \|\dot{\mathbf{x}}_n\|}{\partial \mathbf{p}}^T \tag{10}$$

where $\frac{\partial \|\dot{\mathbf{x}}_n\|}{\partial \mathbf{p}} = \begin{bmatrix} \frac{-a}{\sqrt{(a^2+b^2)^3}} & \frac{-b}{\sqrt{(a^2+b^2)^3}} & 0 \end{bmatrix}$. As will be discussed in Section V-C, the derived uncertainty can be used to reasonably reweight the residual terms that involve normal-flow observations, thereby suppressing the impact of inaccurate normal-flow estimates.

C. Continuous-Time Linear Velocity Estimation

Although the computation of event-based normal flows cannot achieve the frequency as high as the streaming rate of event data (e.g., several million per second), it still can be much higher than the frame rate of a standard camera. Besides, the computation can be carried out in an asynchronous manner,so the resulting event-based normal flows typically do not coincide with the inertial measurements in time (i.e., temporally nonaligned data association). Therefore, we employ a continuous-time formulation, which can bound the size of the optimization problem while enabling data association at any given time. Specifically, we use a cubic B-spline-based parametric model to represent the continuous-time linear velocity as

$$\mathbf{v}^{\mathtt{B}}(u(t)) = \sum_{i=0}^{3} B_{i,3}(u(t)) \mathbf{c}_i \tag{11}$$

where $B_{i,k}$ denotes the basis function, i the index of control points, k (=3) the order of the spline, and $\mathbf{c}_i \in \mathbb{R}^3$ the corresponding control point defined in the space of linear velocity. The function $u(\cdot)$ is a normalization operator, which transfers time t to the spline's parameter domain by means of basis translation [70].

Our goal is to fit a cubic B-spline that simultaneously satisfies the following two criteria. 1) The predicted normal flows are maximally consistent with the asynchronous normal flow measurements. 2) The velocity changes on the spline complies maximally with the IMU preintegration results. The first criterion can be simply evaluated using (4). The second criterion is evaluated, from another equivalent perspective, by calculating the difference between the predicted velocity increment and the preintegration of acceleration in a local frame.

Let us denote the raw accelerometer and gyroscope measurements, $\tilde{\mathbf{a}}_t$ and $\tilde{\boldsymbol{\omega}}_t$, in the body frame at time t by

$$\begin{aligned} \tilde{\mathbf{a}}_t &= \mathbf{a}_t + \mathbf{b}_{a_t} + \mathtt{R}_{\mathtt{W}}^{t} \mathbf{g}^{\mathtt{W}} + \boldsymbol{\sigma}_a \\ \tilde{\boldsymbol{\omega}}_t &= \boldsymbol{\omega}_t + \mathbf{b}_{\omega_t} + \boldsymbol{\sigma}_w \end{aligned} \tag{12}$$

where $\mathbf{b}_a$ and $\mathbf{b}_\omega$ are the accelerometer and gyroscope biases, while $\boldsymbol{\sigma}_a$ and $\boldsymbol{\sigma}_w$ the corresponding additive noise. Let's further consider the IMU preintegration [71] during the time interval $[t_i, t_{i+1}]$. Assuming the IMU biases are known, we integrate the inertial measurements in local frame $\mathtt{B}_i$ (the body frame at time t_i) as

$$\begin{aligned} \tilde{\boldsymbol{\beta}}_{\mathtt{B}_{i+1}}^{\mathtt{B}_i} &= \int_{t_i}^{t_{i+1}} \mathtt{R}_t^{\mathtt{B}_i} (\tilde{\mathbf{a}}_t - \mathbf{b}_{a_t} - \boldsymbol{\sigma}_a) dt \\ \tilde{\boldsymbol{\gamma}}_{\mathtt{B}_{i+1}}^{\mathtt{B}_i} &= \int_{t_i}^{t_{i+1}} \frac{1}{2} \boldsymbol{\Omega}(\tilde{\boldsymbol{\omega}}_t - \mathbf{b}_{\omega_t} - \boldsymbol{\sigma}_w) \boldsymbol{\gamma}_t^{\mathtt{B}_i} dt \end{aligned} \tag{13}$$

where $\boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} -[\boldsymbol{\omega}]_\times & \boldsymbol{\omega} \\ -\boldsymbol{\omega}^{\mathtt{T}} & 0 \end{bmatrix}$. Then, the velocity and orientation can be propagated from t_i to t_{i+1} in the world frame by

$$\begin{aligned} \mathbf{v}_{\mathtt{B}_{i+1}}^{\mathtt{W}} &= \mathbf{v}_{\mathtt{B}_i}^{\mathtt{W}} - \mathbf{g}^{\mathtt{W}} \Delta t_i + \mathtt{R}_{\mathtt{W}}^{\mathtt{B}_i} \boldsymbol{\beta}_{\mathtt{B}_{i+1}}^{\mathtt{B}_i} \\ \mathbf{q}_{\mathtt{B}_{i+1}}^{\mathtt{W}} &= \mathbf{q}_{\mathtt{B}_i}^{\mathtt{W}} \otimes \boldsymbol{\gamma}_{\mathtt{B}_{i+1}}^{\mathtt{B}_i} \end{aligned} \tag{14}$$

where $\mathbf{q}$ refers to the orientation in the form of quaternion. Consequently, the predicted velocity increment during the time interval can be calculated as

$$\hat{\boldsymbol{\beta}}_{t_{i+1}}^{t_i} = \mathtt{R}(\tilde{\boldsymbol{\gamma}}_{t_{i+1}}^{t_i}) \mathbf{v}_{\mathtt{B}_{i+1}}^{\mathtt{B}_{i+1}} + \mathbf{g}^{\mathtt{B}_i} \Delta t_i - \mathbf{v}_{\mathtt{B}_i}^{\mathtt{B}_i}. \tag{15}$$

The detailed derivation of (15) can be found in Appendix B.

Now let us discuss the formulation of the continuous-time state estimation problem. The full state vector is defined as

$$\begin{aligned} \mathcal{X} &= [\mathbf{c}_1, \mathbf{c}_2, \cdots \mathbf{c}_n, \mathbf{b}_1, \mathbf{b}_2, \cdots \mathbf{b}_{n-3}] \\ \mathbf{b}_k &= [\mathbf{b}_{a_k}, \mathbf{b}_{\omega_k}], k \in [1, n-3] \end{aligned} \tag{16}$$

where n denotes the total number of control points, and $\mathbf{b}_k$ the IMU bias in the $k_{\mathtt{th}}$ B-spline segment. Note that there are $n-3$ cubic B-spline segments in the optimization. Since each segment of the B-spline is short, we simply assume the biases are constant within each segment.

Combining (4), (13), and (15), we finally create the following objective function:

$$\mathcal{X}^\star = \arg\min_{\mathcal{X}} \mathcal{C} \tag{17}$$

where the cost is defined as

$$\mathcal{C} \doteq \sum_{t \in \mathcal{S}} \left\| \|\dot{\mathbf{x}}_n(t)\| - \frac{\mathbf{n}^{\mathtt{T}} \mathbf{A}(\mathbf{x}) \mathbf{v}^{\mathtt{B}}(t)}{Z_t(\mathbf{x})} - \mathbf{n}^{\mathtt{T}} \mathbf{B}(\mathbf{x}) \boldsymbol{\omega}^{\mathtt{B}}(t) \right\|_{\sigma_t^c}^2$$

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$$+\sum_{i}^{M}\|\tilde{\boldsymbol{\beta}}_{t_{i+1}}^{t_i}-\mathrm{R}(\tilde{\boldsymbol{\gamma}}_{t_{i+1}}^{t_i})\mathbf{v}_{\mathrm{B}_{i+1}}^{\mathrm{B}_{i+1}}-\hat{\mathbf{g}}^{\mathrm{B}_i}\Delta t_i+\mathbf{v}_{\mathrm{B}_i}^{\mathrm{B}_i}\|^2_{\mathbf{P}_{i+1}^{i}}$$
$$+\sum_{k=2}^{n-3}\|\mathbf{b}_{a_k}-\mathbf{b}_{a_{k-1}}\|^2_{\mathbf{P}_{a_{k-1}}^{a_k}}+\|\mathbf{b}_{\omega_k}-\mathbf{b}_{\omega_{k-1}}\|^2_{\mathbf{P}_{\omega_{k-1}}^{\omega_k}}. \tag{18}$$

The first two terms in (18) implement the abovementioned two criteria, where $\mathcal{S}$ denotes the timestamp set of all involved normal flow estimates, and M the number of time intervals for IMU preintegration. To obtain a maximum a posteriori estimate, we apply the Mahalanobis norm to every residual term. As mentioned in Section V-B, the derived variance of normal-flow magnitude (denoted by σ_t^c) is used to weight each normal-flow term. Intuitively, unreliable normal flow observations usually have a larger variance, thus lower weights are assigned to them. The covariance of the IMU measurements $\mathbf{P}_{i+1}^{i}$ is derived from the dynamics of error terms in (13) as performed in [72]. Note that the orientation is not included in state vector $\mathcal{X}$. Thus, we simply propagate the orientation by integrating raw angular velocities, given the initial orientation. The gravity in the body frame ($\hat{\mathbf{g}}^{\mathrm{B}_i}$) can be constantly estimated using the body-frame orientation via $\hat{\mathbf{g}}^{\mathrm{B}_i}=\mathrm{R}_{\mathrm{W}}^{\mathrm{B}_i}\mathbf{g}^{\mathrm{W}}$. The third and forth terms in (18) are used to suppress the IMU's drift, where $\mathbf{P}_{a_{k-1}}^{a_k}$ and $\mathbf{P}_{\omega_{k-1}}^{\omega_k}$ denote the covariance of the IMU bias.

D. Initialization

To initialize the nonlinear optimization problem [see (18)], we propose a linear way to coarsely determine the instantaneous linear velocity. Given as input the calculated normal flows,[3] their corresponding depth information, and the corresponding measurements of instantaneous angular velocity, a linear system with linear velocity ${}^{\mathrm{B}}\mathbf{v}$ as unknown can be established based on (4):

$$\begin{bmatrix}\mathbf{n}_1^{\mathrm{T}}\mathbf{A}_1\\ \vdots\\ \mathbf{n}_K^{\mathrm{T}}\mathbf{A}_K\end{bmatrix}{}^{\mathrm{B}}\mathbf{v}=\begin{bmatrix}Z_1(\|\dot{\mathbf{x}}_{n,1}\|-\mathbf{n}_1^{\mathrm{T}}\mathbf{B}_1{}^{\mathrm{B}}\boldsymbol{\omega})\\ \vdots\\ Z_K(\|\dot{\mathbf{x}}_{n,K}\|-\mathbf{n}_K^{\mathrm{T}}\mathbf{B}_K{}^{\mathrm{B}}\boldsymbol{\omega})\end{bmatrix}. \tag{19}$$

A minimal solver of (19) requires three observations, and we use RANSAC [73] for robust estimation.

E. Marginalization

To ensure that the event-IMU fusion is computationally tractable, we employ a sliding-window scheme that maintains a fixed number of control points (10 in our implementation). To this end, when a new control point is added into the sliding window, we marginalize out the oldest one together with corresponding IMU measurements and normal-flow observations.[4] As illustrated in Fig. 5, the marginalization preserves

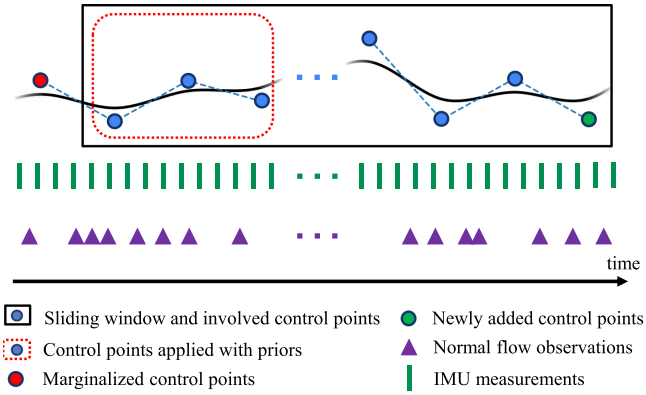


Fig. 5. Factor graph representation of the optimization problem in our system, illustrating the structure of factors and variables, as well as the marginalization strategy.

prior information and applies it to three adjacent control points. This is done by using the *Schur Complement* [75], where a new prior is constructed based on the marginalized state and corresponding measurements, and then applied to the existing prior. Consequently, the cost function in (18) is augmented with a prior term

$$\mathcal{C}'\doteq\mathcal{C}+\|\mathbf{r}_p-\mathbf{H}_p\delta\mathcal{X}_\lambda\|^2 \tag{20}$$

where $\{\mathbf{r}_p,\mathbf{H}_p\}$ refers to the prior information from marginalization, and $\delta\mathcal{X}_\lambda$ denotes the change of $\mathcal{X}_\lambda$ from the prior marginalization point. To perform marginalization, we partition the state variables into three distinct parts: the states to marginalize ($\mathcal{X}_\mu$), their conditionally dependent counterparts connected through error constraints ($\mathcal{X}_\lambda$), and the remaining states ($\mathcal{X}_\gamma$). By leveraging the conditional independence between these three parts, the marginalization process can be streamlined by focusing specifically on the subproblem involving $\mathcal{X}_\mu$ and $\mathcal{X}_\lambda$ [76]

$$\begin{bmatrix}\mathbf{H}_{\mu\mu}&\mathbf{H}_{\mu\lambda}\\ \mathbf{H}_{\lambda\mu}&\mathbf{H}_{\lambda\lambda}\end{bmatrix}\begin{bmatrix}\delta\mathcal{X}_\mu\\ \delta\mathcal{X}_\lambda\end{bmatrix}=\begin{bmatrix}\mathbf{r}_\mu\\ \mathbf{r}_\lambda\end{bmatrix}. \tag{21}$$

Applying the *Schur Complement* yields

$$\mathbf{H}_p\doteq\mathbf{H}_{\lambda\lambda}-\mathbf{H}_{\lambda\mu}\mathbf{H}_{\mu\mu}^{-1}\mathbf{H}_{\mu\lambda} \tag{22}$$
$$\mathbf{r}_p\doteq\mathbf{r}_\lambda-\mathbf{H}_{\lambda\mu}\mathbf{H}_{\mu\mu}^{-1}\mathbf{r}_\mu. \tag{23}$$

VI. Experiments

In this section, we first disclose some implementation details of our pipeline (see Section VI-A). Second, we introduce the experimental setup, including the platforms/configurations used in data generation and collection, and also the characteristics of the data (see Section VI-B). Next, we present both qualitative and quantitative evaluation results, along with comparisons against alternative solutions, using both self-collected (see Section VI-C) and public datasets (see Section VI-D). Moreover, we provide an ablation study that compares the performance of using different configurations and shows the benefits brought by the improvements upon our previous work [21] (see

[3] Note that the normal flows used here are those calculated within a short time interval. In other words, their timestamps can be deemed identical.

[4] The corresponding measurements refer to those within the region defined by the leftmost two knots in the corresponding nonzero domain of the marginalized control point [74].

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Section VI-E). Finally, we analyze the computational performance (see Section VI-F) and discuss the limitation and inspiration of this work (see Section VI-G).

A. Implementation Details

As discussed in Section III, the front-end employs a time-surface renderer and computes the event-based normal flow and corresponding depth at selected events. Particularly, the time surface is updated as in [5], and the event-based normal flow is computed according to (6). To achieve real-time performance, the spatio-temporal plane fitting operation is conducted only on a small portion of events (i.e., those determined via the event selection), while the rest are culled according to the following rules.

1) Events whose pixel coordinates are close (smaller than 5 pixel) to the image plane's boundary are discarded.
2) Events with no more than 15 neighboring data points in the local spatio-temporal volume defined by a 5×5 patch centered at the event's pixel coordinate are discarded.
3) Events whose timestamp differs from the average timestamp of the neighboring events by more than a threshold (5% of the event batch's duration) are discarded.

In this way, only 1% –2% of the events are selected for subsequent normal flow estimation. We observe that obtaining 500 valid normal flow results from 2 k selected events approximately takes 5 ms. Note that our implementation supports controlling the size of the event batch in two ways: 1) Using a constant number of events to adapt to the event streaming rate; 2) Using a constant-length time window to ensure high-frequency execution. In our implementation, the normal flow computation is triggered by the occurrence of every 45 k–75k events depending on the texture richness of the scene. The corresponding depth information is computed as mentioned in Section IV-C with a maximum disparity of 48 pixels and a block size of 17×17 pixels.

The back-end implements the continuous-time nonlinear optimization pipeline using a cubic B-spline. Specifically, a uniform rational B-spline is employed, and we set the knot interval to 0.1 s. The IMU pre-integration interval is set to 0.03 s. Ceres Solver [77] is used to solve the nonlinear optimization problem [see (17)].

B. Experimental Setup and Datasets

To verify the effectiveness of our method in high-speed maneuvers, we create a qualified dataset that consists of both synthetic and real data. In the synthetic dataset, we generate several sequences featuring aggressive maneuvers of a drone using the ESIM simulator [78]. Two synthetic scenes are visualized in Fig. 6. The first scene is a circular corridor in which the drone undergoes a sharp turn. The second one features a drone zipping through randomly distributed boxes. The synthetic drone is equipped rigidly with a stereo event-based camera, a stereo standard camera, and an IMU. Both of the event cameras and standard cameras share identical intrinsic and extrinsic parameters, simulating a pair of DAVIS [79] sensors. The spatial resolution of all the cameras is 346×260 pixel. The standard cameras' exposure time is set to 10 ms, with a frame rate of 30 Hz. Note that the standard cameras are only used by the comparative approaches in Section VI-C. Regarding the configuration of the event generation model, we keep the setting as default. The eps value is 0.001 and the contrast threshold is 0.5. The IMU runs at 200 Hz. The standard deviations associated to the white noise of the accelerometer and gyroscope are set to 1.86×10^{-2} m/s^2 and 1.86×10^{-3} rad/s, and the standard deviations of the accelerometer and gyroscope bias are set to 4.33×10^{-3} m/s^2 and 2.66×10^{-4} rad/s, respectively.

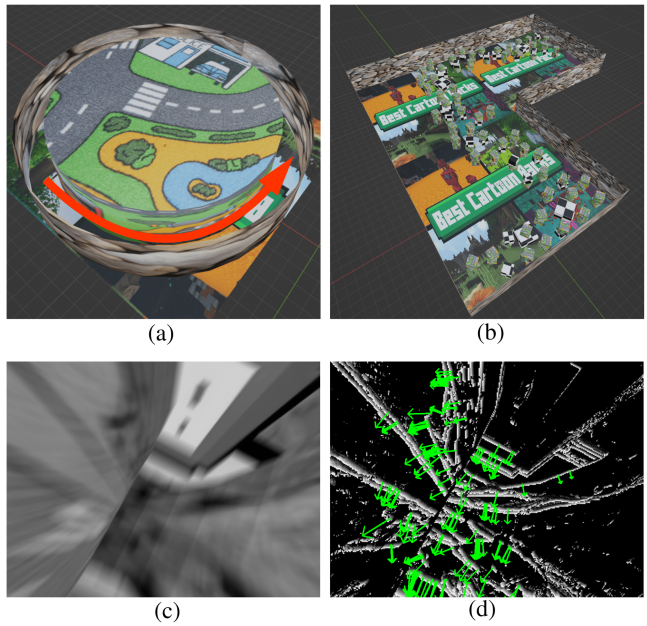


Fig. 6. *Simulation setup for the generation of synthetic data.* Two synthetic scenes are illustrated in (a) and (b). An example of image and event data (represented as a 2-D map via the accumulation of events) from the *circular_corridor* sequences are shown in (c) and (d), respectively. The green arrows in (d) represent the normal flow estimates. (a) Circular corridor. (b) Randomly distributed boxes. (c) Blurry image. (d) Events and normal flow.

We also collect our own real data due to the absence of publicly available datasets[5] that feature aggressive ego motion of a stereo event camera plus an IMU. To comprehensively evaluate our method across diverse motion patterns, we collect data using the following three different setups (see Fig. 7).

1) *Tethered-spinning:* Similar to the data used in [7], we tether the stereo event camera to the end of a rope and spin it. The stereo event camera is spun horizontally, vertically, and in a figure-eight shape, respectively, resulting in three sequences. An example of the spinning motion is shown in Fig. 1(a). The data are collected in a 9 m $\times$ 6 m $\times$ 3.5 m room with randomly distributed boxes. The person spinning the stereo event camera stands in the center of the room.
2) *250 mm quadcoptor:* We construct a 250 mm quadcoptor according to the open-source scheme *Uniquad* [81]. The quadcoptor can perform aggressive maneuvers along predefined trajectories in 3-D space. An illustration of

[5] Note that the aggressive data provided in [80] are captured with a monocular event camera, and thus, not applicable to our task.

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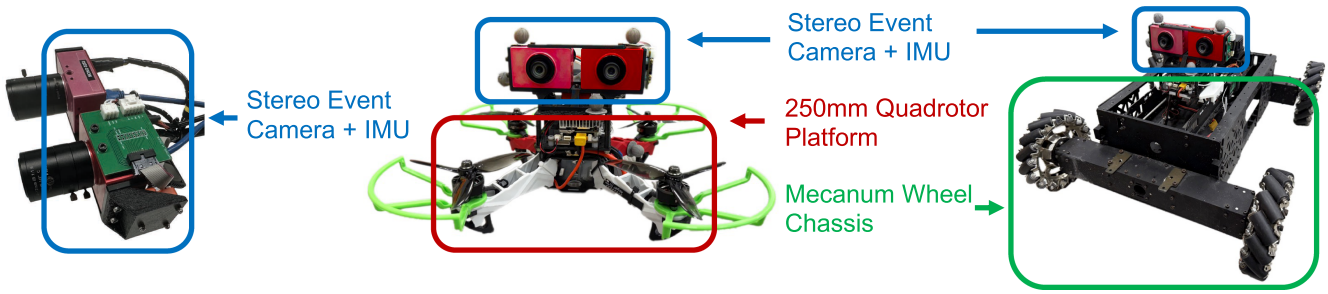


Fig. 7. *Sensors and mobile robot systems used in real-world data collection.* The sensor suite consists of a stereo DAVIS-346 camera and an IMU. From left to right: the sensor suite tethered to a rope, carried by a 250 mm quadcopter, and mounted on a Mecanum-wheel robot, respectively.

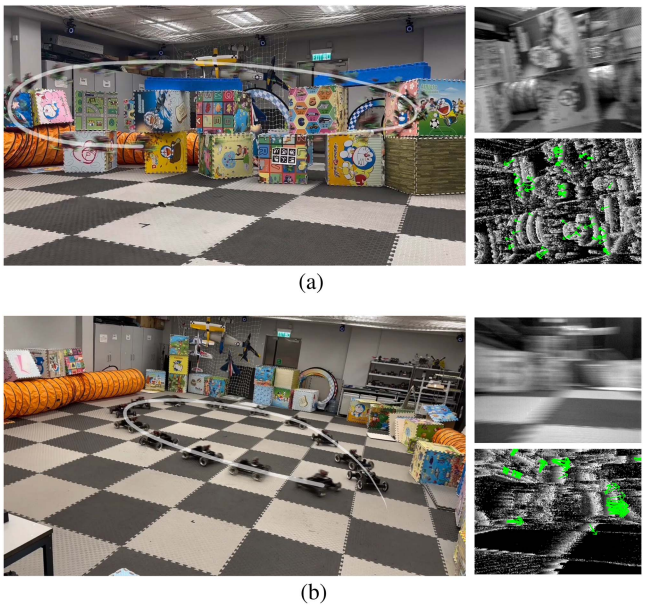


Fig. 8. *Collection of real data.* In each subfigure, the left one illustrates the scene and the mobile platform used for data collection, while the right two show the selected blurry image and corresponding event data (represented as a 2-D map via the accumulation of events). (a) Illustration of the *Drone* subset. (b) Illustration of the *MWR* subset.

the flying quadcopter together with examples of data are given in Fig. 8(a). In the resulting three sequences, the quadcopter undergoes different flying patterns, such as elliptical orbiting and horizontal figure-eight orbiting (with and without the yaw component being fixed).

3) *Mecanum-wheel robot:* We build a holonomic robot system with a Mecanum-wheel chassis, which allows for omnidirectional movement on the ground. In the collected sequences, aggressive translation and yaw variation are simultaneously witnessed, resulting in a 2-D drifting motion pattern as shown in Fig. 8(b).

A pair of DAVIS-346 cameras are used in data collection. All the real data are collected in a room equipped with a motion capture (MoCap) system. By attaching several visual markers to the stereo rig and the carrier, the MoCap system can report 6-DoF poses ($\mathbf{R}_{\mathtt{WM}}$ and $\mathbf{t}_{\mathtt{WM}}$) of the marker coordinate system up to 200 Hz with submillimeter accuracy. Differentiating the position information with respect to time, we can obtain the linear velocity in the world coordinate system (i.e., a MoCap-specific reference coordinate system). With known transformation between the markers and the left camera, we are able to obtain the velocity in the left camera's coordinate system by ${}^{\mathtt{B}}\mathbf{v} = \mathbf{R}_{\mathtt{BM}}\mathbf{R}_{\mathtt{MW}}{}^{\mathtt{W}}\mathbf{v}$. Data smoothening may be required to obtain ground truth velocity at the exact moment the estimation is conducted. All experimental setup and corresponding kinematic characteristics of the sequences are summarized in Table II.

The hand–eye calibration and temporal synchronization procedures for the camera, IMU, and MoCap system follow an approach similar to that described in [80]. Since the DAVIS sensor can simultaneously provide both image and event data (namely, the two modalities share the same intrinsics), we utilize the images directly for camera calibration. The calibration process, including hand–eye calibration and time offset estimation between the camera and IMU, is performed using Kalibr [82]. Similarly, the calibration between the IMU and the mocap system, also involving hand–eye calibration and time offset estimation, is conducted using Basalt [83].

C. Evaluation on Self-Collected Dataset

To justify the effectiveness of our method, we compare it against several alternative approaches capable of estimating linear velocity. Most of these methods take event data (sometimes combined with frames) as input and output trajectory and/or velocity estimates. The first one is IMU integration, which only requires the initial orientation and initial linear velocity as a prior. The second one is the VINS-Fusion pipeline [84], which is widely applied in state estimation tasks of small unmanned aerial vehicles. To make a fair comparison, VINS-Fusion is evaluated after being successfully initialized. The third one, called Ultimate SLAM (USLAM) [39], is an open-source visual-inertial odometry method using as input the frames (F), events (E), and measurements of an IMU. By blocking the image channel, USLAM can be regarded as an event-inertial fusion pipeline like [7]. Both configurations, denoted by USLAM(E+F) and USLAM(E), are used in the comparison. The fourth one is ESVIO [9], a feature-based VIO system using as input the frames and events in a stereo configuration. The image channel can also be blocked, resulting in a stereo event-inertial pipeline (denoted by ESIO). The last one is ESVO2 [33], a recent state-of-the-art direct method for stereo event cameras. It features accurate trajectory estimation in large-scale environments and efficient

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TABLE II
EXPERIMENTAL SETUP AND CHARACTERISTICS OF OUR REAL DATA

Seq. Name	Setup	Scene	Motion Pattern	Max Linear Vel.	Max Angular Vel.
Synthetic_1	Drone simulation	Circular corridor	Sharp turn	7.63 m/s	2.81 rad/s
Synthetic_2	Drone simulation	Circular corridor	Sharp turn	15.24 m/s	8.98 rad/s
Synthetic_3	Drone simulation	Circular corridor	Sharp turn	25.28 m/s	15.43 rad/s
Synthetic_4	Drone simulation	Randomly distributed boxes	Zip through obstacles	3.83 m/s	5.56 rad/s
Synthetic_5	Drone simulation	Randomly distributed boxes	Zip through obstacles	4.32 m/s	8.20 rad/s
Spinning_1	Tethered spinning	Flight arena	Horizontal spin	5.50 m/s	13.77 rad/s
Spinning_2	Tethered spinning	Flight arena	Vertical spin	6.50 m/s	22.03 rad/s
Spinning_3	Tethered spinning	Flight arena	Figure-8 spin	7.60 m/s	17.97 rad/s
Drone_1	250 mm quadcopter	Flight arena	Elliptical orbiting (w/ yaw)	4.36 m/s	3.73 rad/s
Drone_2	250 mm quadcopter	Flight arena	Horizontal figure-8 orbiting (w/o yaw)	4.35 m/s	3.09 rad/s
Drone_3	250 mm quadcopter	Flight arena	Horizontal figure-8 orbiting (w/ yaw)	4.31 m/s	4.71 rad/s
MWR_1	Mecanum-wheel robot	Flight arena	Planar drifting	3.90 m/s	11.98 rad/s
MWR_2	Mecanum-wheel robot	Flight arena	Planar drifting	4.69 m/s	12.82 rad/s
MWR_3	Mecanum-wheel robot	Flight arena	Planar drifting	4.69 m/s	12.30 rad/s
MWR_4	Mecanum-wheel robot	Flight arena	Planar drifting	4.18 m/s	12.28 rad/s

TABLE III
MEAN AND STANDARD DEVIATION OF AVE (M/S) ON SYNTHETIC DATA

Seq. Name	IMU	VINS-Fusion [84]	USLAM(E) [39]	USLAM(E + F) [39]	ESIO [9]	ESVIO [9]	ESVO2 [33]	Ours
	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)
Synthetic_1	2.67 (1.83)	0.48 (0.28)	2.92 (1.48)	0.21 (0.54)	0.13 (0.06)	**0.10** (0.05)	0.36 (0.58)	**0.10** (**0.04**)
Synthetic_2	5.73 (3.67)	0.87 (0.19)	19.59 (13.45)	0.24 (0.70)	0.18 (0.14)	0.16 (0.11)	2.34 (1.50)	**0.14** (**0.05**)
Synthetic_3	7.83 (4.17)	19.80 (6.33)	172.58 (104.57)	15.72 (9.16)	1.30 (0.26)	0.81 (0.28)	2.63 (1.86)	**0.23** (**0.10**)
Synthetic_4	0.54 (0.05)	0.14 (0.12)	0.45 (0.41)	0.06 (0.07)	0.13 (0.09)	0.06 (**0.03**)	1.29 (1.75)	**0.05** (**0.03**)
Synthetic_5	0.63 (0.26)	0.23 (0.13)	0.48 (0.38)	0.08 (0.05)	0.19 (0.09)	0.15 (0.11)	0.43 (0.33)	**0.07** (**0.04**)

mapping that scales well with modern high-definition event cameras.

Evaluation metrics: To quantitatively evaluate the velocity estimates, we apply as evaluation metrics the absolute velocity error (AVE) and the relative velocity error (RVE). The latter is defined as the ratio of the AVE and the magnitude of ground truth velocity, namely

$$\text{RVE} = \frac{\|\mathbf{v}_{gt} - \mathbf{v}_{est}\|_2}{\|\mathbf{v}_{gt}\|_2} \times 100\%. \tag{24}$$

In addition, the standard deviation of AVE is also used to assess the robustness of algorithms.

1) Evaluation on Synthetic Data: We evaluate all abovementioned methods using the synthetic sequences. Detailed AVE results can be found in Table III, in which best results are highlighted in bold. To have a close-up view of each method's performance, we select as representatives the results of two sequences that feature the most aggressive maneuvers in each scene.

As illustrated in Fig. 9, obvious accumulated errors are witnessed in the velocity estimates from the IMU-integration method. VINS-Fusion performs normally on *Synthetic_5* but comes across a failure of camera pose tracking on *Synthetic_3* because of blurred frames induced by aggressive maneuvers. USLAM assumes that the sensor remains static during the initialization phase. To make a fair comparison, we provide USLAM with additional static-state data at the beginning to ensure successful initialization. Compared to the fairly good performance on *Synthetic_5*, USLAM struggles on *Synthetic_3* and reports severely large AVE. This is mainly caused by the theoretically-flawed approach for event-based feature tracking, which fails when the assumption of small variations in local depth is violated. In comparison, USLAM(E+F) performs much better because of additionally incorporating frames that enhance feature matching. Still, USLAM(E+F) encounters failures of camera pose tracking on *Synthetic_3*, similar to VINS-Fusion. Further improvements are witnessed when building on top of the feature-based method with a stereo configuration. In detail, ESIO delivers reliable velocity estimates, and ESVIO achieves the best among all feature-based method. Compared to the monocular counterparts, ESIO and ESVIO adopt a more cautious feature matching scheme that undergoes a spatial–temporal verification. This additional spatial verification, allowed by the stereo configuration, alleviates the theoretical flaws in the method for event-based feature tracking, thus leading to better performance. However, the AVE of ESVIO's reslult on *Synthetic_3* is still unsatisfactory, indicating the limitations of existing feature-based methods. Unsurprisingly, ESVO2, as a representative direct method, struggles with high-speed maneuvers due to untimely updates of the local map, particularly under

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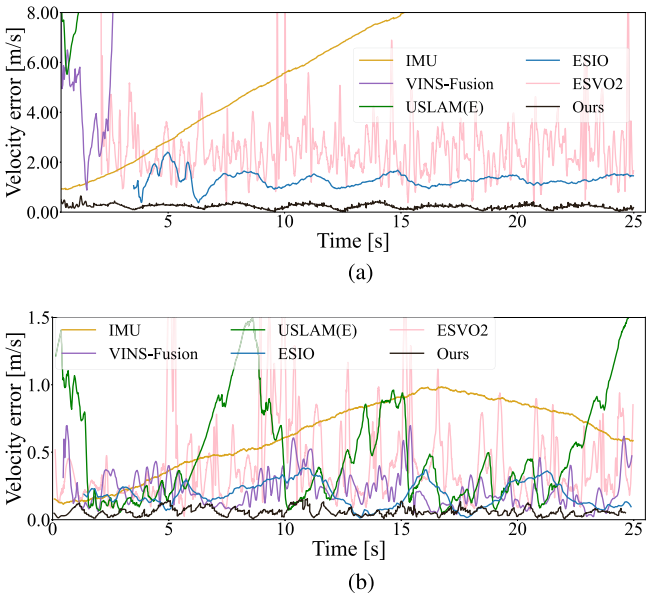


Fig. 9. *AVE results of two representative sequences.* Note that the AVE results of *VINS-Fusion* and *USLAM(E)* in (a) are too large to be visualized inside the plot. (a) AVE results on *Synthetic_3*. (b) AVE results on *Synthetic_5*.

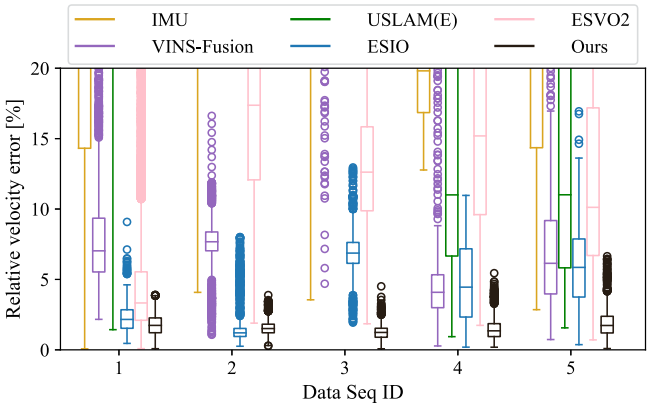


Fig. 10. RVE results on our synthetic dataset. Note that the RVE of *IMU* and *USLAM(E)*, and the RVE of *VINS-Fusion* on *Synthetic_3* is too large to be visualized inside the plot.

large angular velocities (e.g., *Synthetic_2* and *Synthetic_3*). To run ESVO2 on the synthetic sequences, we actually reduce the playback rate by a factor of 0.1, allowing the mapping thread to catch up. However, the high angular velocity in these sequences reduces overlap between successive observations, limiting depth fusion operations. This degrades the local map quality and, consequently, tracking performance. In comparison, our method always gives the best results in terms of both AVEs and their standard deviation, as consistently seen in Table III.

Correspondingly, the RVE results are illustrated using box plots in Fig. 10. Compared to other methods, the RVE statistics of our method exhibit a more compact distribution with lower median RVE and fewer outliers. In conclusion, our method achieves the best performance in terms of AVE and RVE on the synthetic dataset.

2) Evaluation on Real Data: We also evaluate all methods using the self-collected real data, which consist of three subsets including "Spinning," "Drone," and "MW," respectively. The *Spinning* subset exhibits the most challenging scenarios featuring the highest dynamics among the three. The *Drone* and *MWR* subsets also capture aggressive ego motion. To do this, the quadcopter and Mecanum-wheel robot have been actuated by almost approaching the upper bound of their mobility during data collection. The quantitative evaluation results (using AVE) are listed in Table IV, where the best results are highlighted in bold.

The conclusion is consistent with that drawn in the evaluation on synthetic data. As seen in Table IV, the IMU-integration method, used as the baseline, always gives the worst result. VINS-Fusion is slightly better, but easily fails due to motion blur effects induced by aggressive ego motion. Enhanced by additional feature matching on event data, USLAM and ESVIO perform better than VINS-Fusion, but their AVEs are still too high for reliable state estimation. The primary reason for these large AVEs lies in the unreliable feature matching under aggressive ego motion. The rapid camera movements make it difficult to establish consistent temporal data association on either images or events. Rapid scene changes cause significant interframe feature displacements, bringing great challenges to feature tracking algorithms. As for event-based feature tracking, the employed ad hoc feature tracking method on motion-compensated event map cannot always guarantee reliable feature correspondences. This is due to the motion-dependent nature of event data, which results in a lack of guaranteed appearance consistency in motion-compensated event maps. Corner-like features (e.g., junctions) may vanish in the presence of sudden velocity changes. Furthermore, the motion-compensation operation employed in USLAM and ESVIO relies on a constant-depth assumption within the local map (for the purpose of efficiency), which can lead to imperfect event warping and, as a result, less sharp event maps that are suboptimal for feature detection and tracking. The results of ESVO2 are consistent with those on the synthetic data. It typically fails in the presence of large angular velocities (as observed in the *Spinning* and *MWR* subsets) but performs reasonably good when high-speed translational motion dominates. According to our investigation, real stereo data exhibits poorer spatio-temporal consistency under high angular velocities compared to synthetic sequences. This explains ESVO2's failure on the *Spinning* and *MWR* subsets. In comparison, our method always gives the best results (both AVEs and their standard deviation). This is because our method is based on instantaneous normal-flow observations, which finds a way around the explicit data association problem on event data, and therefore aligns better with the differential working principle of event cameras.

We also choose some sequences from real data as representatives to provide a close-up view of each method's performance, as shown in Fig. 11. Note that the motion in *MWR_2* is 2-D, and therefore the Y component (in the body frame) of the velocity estimates is close to zero. In general, the results of our method are always in better agreement with the ground truth velocities, especially at the peaks, which indicate that a sudden change of motion has occurred. This behavior indirectly justifies that our method adapts better to the motion-dependent nature of event data than the feature based methods.

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TABLE IV
MEAN AND STANDARD DEVIATION OF AVE (M/S) ON SELF-COLLECTED REAL DATA

Seq. Name	IMU	VINS-Fusion [84]	USLAM(E) [39]	USLAM(E + F) [39]	ESIO [9]	ESVIO [9]	ESVO2 [33]	Ours
	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)	AVE (std)
Spinning_1	5.47 (1.31)	3.57 (2.38)	1.08 (0.80)	1.14 (0.58)	1.08 (0.56)	1.02 (**0.51**)	Failed	**0.96** (0.65)
Spinning_2	9.21 (8.06)	8.43 (7.84)	6.31 (4.44)	5.37 (2.04)	10.14 (8.80)	8.66 (7.15)	Failed	**0.98** (**0.95**)
Spinning_3	8.89 (7.05)	7.62 (8.70)	9.67 (6.05)	6.09 (5.68)	10.43 (10.38)	10.78 (10.89)	Failed	**1.04** (**0.70**)
Drone_1	13.62 (10.32)	2.14 (1.97)	0.41 (0.17)	0.31 (0.16)	0.41 (0.29)	0.28 (0.15)	0.74 (0.65)	**0.25** (**0.14**)
Drone_2	4.70 (4.72)	0.25 (0.16)	0.27 (0.12)	0.25 (0.15)	0.25 (0.14)	0.25 (0.12)	0.62 (0.55)	**0.22** (**0.10**)
Drone_3	21.53 (21.85)	10.49 (10.46)	0.32 (**0.14**)	**0.30** (**0.14**)	8.54 (8.20)	0.31 (0.16)	1.09 (1.03)	**0.30** (**0.14**)
MWR_1	15.38 (8.59)	4.20 (2.31)	0.53 (0.62)	0.39 (0.28)	0.36 (0.34)	**0.26** (**0.17**)	Failed	0.34 (0.19)
MWR_2	22.32 (13.12)	12.27 (7.05)	0.69 (0.36)	0.41 (**0.19**)	0.51 (0.24)	0.41 (0.27)	Failed	**0.37** (0.23)
MWR_3	12.68 (5.93)	19.79 (10.77)	0.53 (0.28)	0.50 (0.28)	0.59 (0.45)	0.86 (1.04)	Failed	**0.39** (**0.23**)
MWR_4	7.30 (4.86)	7.60 (3.91)	0.45 (0.21)	0.41 (0.27)	16.40 (9.76)	3.38 (2.24)	Failed	**0.34** (**0.19**)

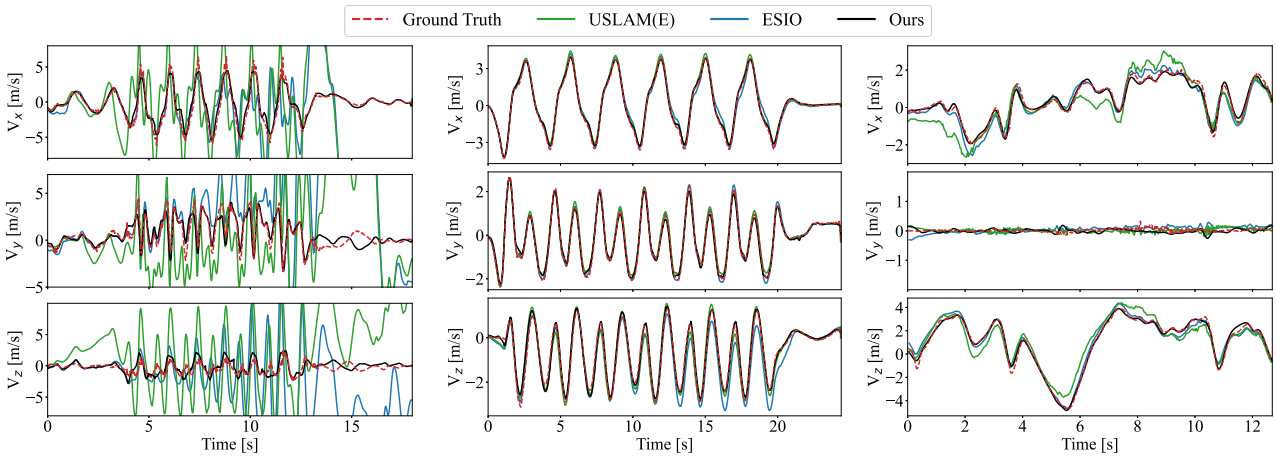


Fig. 11. Estimated linear velocity compared against ground truth and other two counterparts (i.e., ESIO [9] and USLAM(E) [39]) on three sequences of our real data. (a) Results on *Spinning_3*. (b) Results on *Drone_1*. (c) Results on *MWR_2*.

In addition, we perform a qualitative evaluation by illustrating the trajectories recovered using different methods (see Fig. 12). Three sequences (i.e., *Spinning_3*, *Drone_1*, and *MWR_2*) are selected for illustration. VINS-Fusion, USLAM, and ESVIO can directly return camera trajectories. As for the pure IMU-based solution and our method, dead reckoning is carried out. As seen in Fig. 12, our method leads to the most reliable trajectory estimates, and ESIO performs the second best, while the others notably drift away. Compared to the dead-reckoning result of an IMU, our method requires only a single integration. Correspondingly, the quantitative evaluation results, expressed in terms of absolute trajectory error (ATE) [85], are presented in Table V. We demonstrate that the accurate linear velocity recovered by our method leads to better trajectories in terms of global consistency, proving our method as a better solution to the dead-reckoning task under high-speed maneuvers.

TABLE V
ATE (M) OF THE TRAJECTORIES IN FIG. 12

Seq. Name	IMU	VINS-Fusion [84]	USLAM(E) [39]	ESIO [9]	Ours
Spinning_3	50.17	26.00	31.21	41.62	**1.63**
Drone_1	78.99	11.83	1.43	2.21	**0.63**
MWR_2	105.14	56.45	1.71	0.67	**0.26**

The bold formatting is a best-performing results.

D. Evaluation on Public Dataset

We notice the availability of other public datasets [9], [49], [50], [51], [52], [86] featuring event cameras paired with standard cameras and IMUs, with data captured in both indoor and outdoor settings. However, the characteristics of these dataset do not align well with the definition of high-speed maneuvers presented in this article. Most of them, such as MVSEC [49], DSEC [50], and M3ED [52], capture high-speed driving scenarios where the vehicles are positioned far from the scene, leading to small motion parallax and negligible motion blur. The TUM-VIE dataset [86] contains sequences with aggressive camera motion; however, the provided ground truth does not cover the entire sequence, limiting its usability. The VECtor dataset [51] is characterized by a comprehensive sensor

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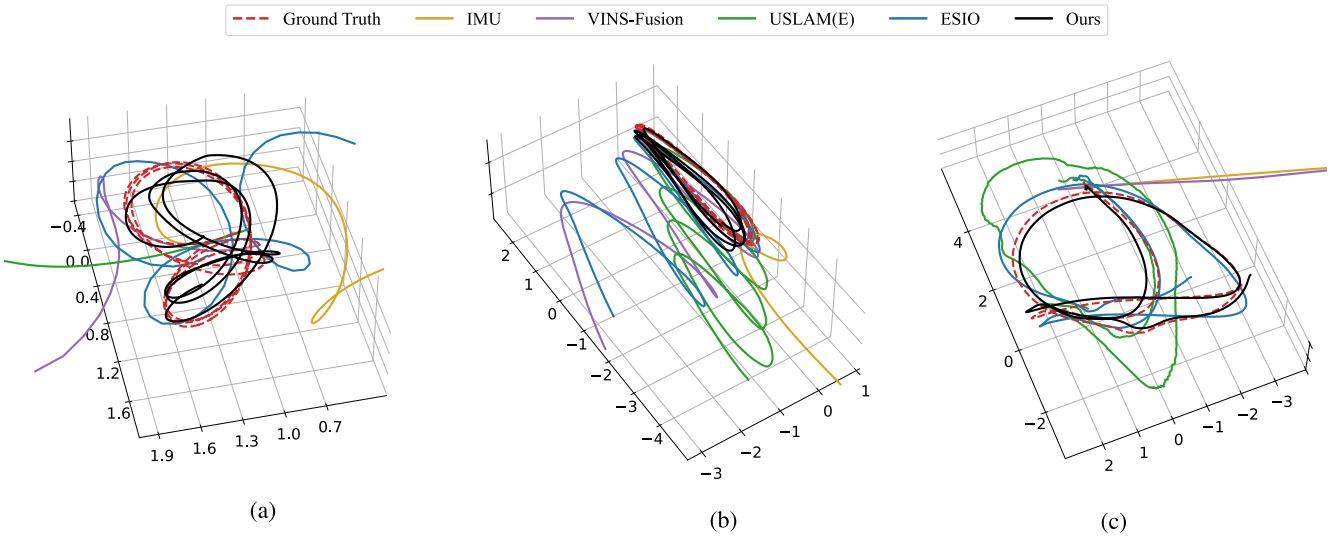


Fig. 12. Dead-reckoning results of the proposed method compared to ground truth and other two counterparts (i.e., ESIO [9] and USLAM(E) [39]) on three sequences of our real data. (a) Results on *Spinning_3*. (b) Results on *Drone_1*. (c) Results on *MWR_2*.

TABLE VI
MEAN AND STANDARD DEVIATION OF AVE (M/S) AND RPE (M/M) ON PUBLIC SEQUENCES

Dataset	Seq. Name	USLAM (E) [39]			ESIO [9]			ESVO2 [33]			Ours		
		AVE	std(σ)	RPE	AVE	std(σ)	RPE	AVE	std(σ)	RPE	AVE	std(σ)	RPE
MVSEC [49]	*Outdoor Night 1*	39.21	33.97	4.13	222.33	162.43	50.53	Failed	Failed	Failed	**1.80**	**1.15**	**0.41**
	Outdoor Night 2	308.89	218.06	23.79	39.25	41.29	8.50	Failed	Failed	Failed	**1.40**	**0.94**	**0.34**
	Outdoor Night 3	74.44	51.81	39.10	88.95	61.29	37.49	Failed	Failed	Failed	**1.51**	**1.08**	**0.35**
TUM-VIE [86]	*mocap-shake*	**0.19**	**0.10**	**0.54**	-	-	-	2.35*	1.71*	1.03*	0.79	0.55	0.61
	mocap-shake2	**0.38**	**0.26**	**0.66**	-	-	-	4.77*	2.55*	1.06*	0.97	0.74	0.76
DSEC [50]	*zurich_city_04_a*	1.05	0.92	0.19	1.58	0.76	0.86	**0.29**	**0.23**	**0.10**	0.48	0.34	**0.10**
	zurich_city_04_b	3.48	1.73	1.83	0.87	0.37	0.49	0.45	0.32	0.14	**0.34**	**0.12**	**0.06**
	zurich_city_11_a	5.11	0.66	6.50	0.34	0.21	0.09	**0.18**	**0.15**	**0.05**	0.22	0.18	**0.05**
	zurich_city_11_b	7.14	4.50	2.72	0.49	0.39	0.72	**0.16**	**0.12**	**0.04**	0.37	0.26	0.11

"-" represents the algorithm encounters an initialization failure. "" represents the algorithm can not finish the entire sequence.*
The bold formatting is a best-performing results.

configuration and precise synchronization across its heterogeneous sensors, but sequences featuring high-speed maneuvers are not seen. While the HKU dataset [9] contains aggressive motion useful for testing, we exclude it due to imprecise extrinsic camera parameters and notable differences in the sensor characteristics (i.e., hardware setup regarding the generative model of event data) between the left and right units of the stereo event camera used. These inconsistencies undermine spatio-temporal coherence, which is critical for stereo event-based methods.

To ensure a comprehensive evaluation, we select multiple sequences from these public datasets [49], [50], [86], encompassing large-scale outdoor environments and nighttime HDR scenes. In addition to the mean and standard deviation of AVE, we also incorporate relative pose error (RPE) [85] to evaluate the translation drift per meter. The results, presented in Table VI, demonstrate that our method generalizes effectively to scenarios beyond our defined scope of high-speed maneuvers. The *outdoor night* sequences in MVSEC [49] depict nighttime driving scenarios, and we select continuous motion segments that exceed 100 s for evaluation. In these sequences, a significant portion of the event data are generated by flickering lights and brightness fluctuations, while the low-light conditions further amplify the noise levels in the event data. These factors severely impact methods, such as USLAM and ESIO, which rely on feature tracking from event-based image representations. Moreover, the stereo camera setup, characterized by a short baseline and low resolution, undermines depth estimation accuracy, further diminishing the performance of ESIO and causing the failure of ESVO2. In contrast, our method leverages local normal flow observations, which remain robust despite event appearance variations. By incorporating depth estimation into the RANSAC-based velocity initialization, our approach mitigates the impact of imprecise depth estimation. As a result, our method is the only one capable of producing reasonable outcomes on these sequences.

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TABLE VII
RESULTS OF THE ABLATION STUDIES

Seq. Name	EVIV [21]		Benosman-NF		w/ NF Uncertainty		w/ Marginalization		Ours	
	AVE	std(σ)	AVE	std(σ)	AVE	std(σ)	AVE	std(σ)	AVE	std(σ)
Spinning_1	1.82	1.25	7.34	2.74	1.27	0.82	1.19	0.77	**0.96**	**0.65**
Spinning_2	1.83	1.68	5.49	4.56	1.10	**0.95**	1.11	**0.95**	**0.98**	**0.95**
Spinning_3	1.65	1.36	6.11	3.67	1.12	0.85	1.12	0.82	**1.04**	**0.70**
Drone_1	0.39	0.21	20.39	9.45	0.28	0.17	0.29	**0.11**	**0.25**	0.14
Drone_2	0.32	0.16	20.23	10.24	0.26	0.14	0.27	0.15	**0.22**	**0.10**
Drone_3	0.53	0.24	32.50	16.32	0.41	0.29	0.36	0.23	**0.30**	**0.14**
MWR_1	1.13	0.93	10.07	6.61	1.07	1.11	0.38	0.23	**0.34**	**0.19**
MWR_2	2.15	1.53	7.99	5.57	2.18	2.13	0.39	0.25	**0.37**	**0.23**
MWR_3	2.00	1.91	6.14	4.32	1.87	1.86	0.44	0.26	**0.39**	**0.23**
MWR_4	1.20	1.23	6.90	4.66	0.96	1.27	0.51	0.27	**0.34**	**0.19**

We also evaluate our method on the *mocap-shake* sequences, the most challenging scenarios in TUM-VIE [86]. ESIO fails to initialize, and ESVO2, for the same reasons outlined in Section VI-C, cannot guarantee a success on the entire sequence. USLAM achieves the best performance on these sequences. According to our investigation, the hardware's limited data transfer rate leads to significant data loss, hindering the formation of coherent planes from sparse local events in the spatio-temporal domain. This issue results in poor normal flow estimation in our approach, ultimately leading to less accurate velocity estimation.

DSEC [50] is a large-scale dataset featuring outdoor driving scenes. While ESIO outperforms USLAM by leveraging enhanced binocular information integration, its effectiveness is still limited by inherent theoretical flaws in the event-based feature matching. Due to the relatively small motion parallax, ESVO2 benefits from a comprehensive mapping and tracking pipeline, achieving the best performance among these methods. In contrast, our approach is specifically tailored for high-speed scenarios as a map-free velocity estimator, resulting in slightly inferior performance compared to ESVO2.

E. Ablation Study

To justify our method for NF computation and also demonstrate the improvements brought by the newly introduced components, including the NF-uncertainty-based reweighting scheme and the marginalization operation, to our previous work (EVIV [21]), we carry out an ablation study using our real data. Specifically, we compare the performance of five different configurations of our method. The first one is our early work (EVIV), which realizes linear velocity estimation via fusing the event-based normal flow and measurements of an IMU. We use it as the baseline. The second one, denoted by "Benosman-NF," refers to the configuration that replaces the method for computing event-based normal flow in the baseline with Benosman's method [30]. The third and fourth configurations are built on top of the baseline, by enabling the NF-uncertainty-based reweighting scheme ("w/ NF Uncertainty") and the marginalization operation ("w/ Marginalization"), respectively. The last one is our complete pipeline ("Ours") with all newly added components enabled.

TABLE VIII
COMPUTATION TIME OF OUR ALGORITHM

	Module	Avg. Time (ms)
Front-end	Time Surface Update	3.97 ($\sim$70 k)
	Event Selection	0.83 ($\sim$70 k)
	NF and Depth Estimation	7.29 ($\sim$2 k)
	Subtotal	12.19
Back-end	Velocity Initialization	0.37 ($\sim$500)
	Nonlinear Optimization	8.61
	Marginalization	0.42
	Subtotal	9.40

The accuracy and robustness performance of each configuration can be found in Table VII, where best results are highlighted in bold. Compared to the baseline, "Benosman-NF" typically gives worse results, justifying our derivation of event-based normal flow computation. In addition, obvious improvements are witnessed when each of the new components is enabled over the baseline. The performance of "w/ NF Uncertainty" is generally on par with that of "w/ Marginalization," except in *MWR* where the latter is notably better. This is mainly due to the relatively worse quality of normal flow estimates observed in *MWR* (due to the textureless ground plane occupying a significant portion of the field of view), resulting in unreliable constraints in the normal-flow term of the continuous-time nonlinear optimization. The marginalization operation, however, can preserve historical information and impose these priors on the eldest control points in the sliding window, thereby suppressing fluctuations in velocity estimates caused by unreliable normal-flow terms. As expected, our complete pipeline ("Ours") almost gives the best result in all sequences, indicating that the joint employment of the two new components can benefit each other.

F. Computational Performance

Our pipeline is implemented in C++ on ROS and all experiments are performed in real time on a desktop PC with an Intel Core 13700 K CPU. The computation time of all the main modules is summarized in Table VIII. The numbers shown inside

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the parentheses represent the amount of data being processed. In our experiment, the time surface is updated by the occurrence of 70 k events, from which 2 k are selected for computing normal flow and depth. We further select 500 normal flow estimates (and corresponding depth) with the smallest fitting error and feed them to the back-end. The subtotal computation time of the front-end is about 12 ms. In the back-end, the velocity initialization, using as input the 500 normal flow and depth estimates, takes about 0.37 ms. The continuous-time nonlinear optimization typically needs no more than 10 iterations to converge, and it takes about 8 ms. The subsequent marginalization operation takes less than 0.5 ms. In subtotal, every update of the back-end takes less than 10 ms.

Note that the normal flow and depth computations are performed simultaneously in two independent threads. Fed with the front-end's output, the back-end is executed on another separated thread in parallel. Consequently, the whole system can output the body-frame linear velocity up to 80 Hz, and the efficiency can be further improved via code optimization.

G. Discussion

1) Limitations: The proposed method has some limitations. First, the event-based normal flow is computed asynchronously in the sense that the computation is triggered by the occurrence of a certain number of events. In other words, the event-based normal flows are computed in an asynchronous batch manner. Therefore, a batch of flow estimates may share the identical timestamps, but the time interval between two successive batches is not a constant. Although the resulting temporal nonalignment between data can be effectively tackled by the proposed continuous-time framework, we need to investigate if a fully asynchronous computation (i.e., an event-by-event manner) will further unlock the potential of event cameras. The way depth information is computed will be adjusted accordingly. Second, the uniform cubic B-spline parameterization would be replicated by a nonuniform one to deal with sudden changes of velocity. This will be investigated as another future work.

2) Why Choose Linear Velocity as the Only Unknowns?: The angular velocity can be definitely included in the state vector as unknowns and solved using our method. However, it will result in worse performance in terms of accuracy and efficiency. First, it is well known that decoupling rotation and translation estimation, if applicable, leads to better accuracy. This is because the ambiguity in visual cues is circumvented when computing the rotation and translation separately. Moreover, the estimated 3-D angular velocity from events, as reported in [22], is just as accurate as the gyroscope's measurements at its best performance. Thus, it is granted to use the angular velocity measurements as input. Second, it is obvious that the involvement of angular velocity in the state vector will complicate the system, and thus, slows down the computation. Compared to the IMU's dead-reckoning process, which requires integrating from acceleration and involves two integration steps, leveraging our accurate linear velocity estimates can eliminate one integration step, thereby significantly reducing the accumulated integration error in the dead-reckoning process. Therefore, we believe that the linear velocity estimation method presented in this article, along with the single-integration dead-reckoning strategy, represents a better solution for state estimation tasks during high-speed maneuvers. This is why we choose linear velocity as the only unknowns.

3) Frame-Based Methods Versus Event-Based Methods: It is also worth thinking about which is the optimal design for handling state estimation under aggressive maneuvers. For standard frame-based cameras, the motion blur can be mitigated by adaptively reducing the integration time. However, the adaptive exposure control complicates the system and is thus not widely applied in the community. As for event-based cameras, they are naturally qualified for high-speed perception tasks due to the high temporal resolution. Besides, the system delay induced by an event-based camera is almost negligible compared to that caused by the inter-frame gap between successive exposures of a standard frame-based camera. Nonetheless, we would like to clarify that although event cameras have a very high dynamic range, event-based methods are not applicable in very dark environments.

VII. Conclusion

We have presented a novel design of state estimator for aggressive maneuvers using a stereo event camera and an IMU. The proposed system is called event-based visual-inertial velometer because the state estimation problem is solved at the level of the first-order kinematics, namely linear velocity. Our front-end takes as input events acquired by a pair of event cameras in stereo configuration, estimating the normal flow and corresponding depth. The event-based normal flow directly communicates the linear velocity, closely aligned with the differential working principle of event cameras. The back-end is built on top of a continuous-time framework, which can handle data associations across heterogeneous and temporally nonaligned measurements. The proposed method has been comprehensively evaluated using self-collected data that feature high-speed maneuvers commonly seen in practical applications of robotics and computer vision. Results on public datasets also demonstrate that our method generalizes effectively to scenarios beyond our defined scope of high-speed maneuvers. Compared with existing visual-event-inertial-based solutions, our method achieves superior performance, proving its suitability for state estimation tasks under high-speed maneuvers.

Appendix A
Derivation of NF Norm [see (9)]

Given the resulting coefficients of the plane (i.e., $t = ax + by + c$) fitted in the spatio-temporal domain, we can obtain the local gradient of the temporal surface

$$\begin{bmatrix} \nabla_x \mathcal{T} \\ \nabla_y \mathcal{T} \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}. \tag{25}$$

By substituting (25) into (5), we have

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{\|\dot{\mathbf{x}}_n\|} \begin{bmatrix} \cos(\alpha) \\ \sin(\alpha) \end{bmatrix} \tag{26}$$

which can be readily derived to $\|\dot{\mathbf{x}}_n\| = \frac{1}{\sqrt{a^2+b^2}}$.

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APPENDIX B
DERIVATION OF PREDICTED VELOCITY INCREMENTS [SEE (15)]

Following the result in (9) of [72], we have

$$\hat{\boldsymbol{\beta}}_{t_{i+1}}^{t_i} = \mathrm{R}_{\mathrm{W}}^{\mathrm{B}_i}\left(\mathrm{R}_{\mathrm{B}_{i+1}}^{\mathrm{W}}\,\mathbf{v}_{\mathrm{B}_{i+1}}^{\mathrm{B}_{i+1}} + \mathbf{g}^{\mathrm{W}}\Delta t_i - \mathrm{R}_{\mathrm{B}_i}^{\mathrm{W}}\,\mathbf{v}_{\mathrm{B}_i}^{\mathrm{B}_i}\right). \tag{27}$$

By incorporating the rotation $\mathrm{R}_{\mathrm{W}}^{\mathrm{B}_i}$ into the parentheses, we have

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{t_{i+1}}^{t_i} &= \mathrm{R}_{\mathrm{B}_{i+1}}^{\mathrm{B}_i}\,\mathbf{v}_{\mathrm{B}_{i+1}}^{\mathrm{B}_{i+1}} + \mathbf{g}^{\mathrm{B}_i}\Delta t_i - \mathbf{v}_{\mathrm{B}_i}^{\mathrm{B}_i} \\ &= \mathrm{R}(\tilde{\boldsymbol{\gamma}}_{t_{i+1}}^{t_i})\,\mathbf{v}_{\mathrm{B}_{i+1}}^{\mathrm{B}_{i+1}} + \mathbf{g}^{\mathrm{B}_i}\Delta t_i - \mathbf{v}_{\mathrm{B}_i}^{\mathrm{B}_i}. \end{aligned}\tag{28}$$

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